HCL Hackathon – 30-Day Customer Spend Prediction

Problem Overview

Retailers want to predict **how much a customer will spend in the next 30 days** based on their past purchasing behaviour.

Predicting short-term customer spend helps businesses:

- Identify high-value customers
- Target marketing campaigns effectively
- Improve customer retention
- Optimize promotional strategies

Machine Learning Problem

Supervised Learning – Regression

Input Features (X):

Customer behaviour before a cutoff date:

- Purchase frequency
- Total past spending
- Days since last purchase
- Product purchasing patterns
- Customer country

Target Variable (y):

Total amount a customer spends **in the next 30 days**.

This represents **Short-Term Customer Lifetime Value (30-Day CLV)**.

Dataset Selection

Brazilian E-Commerce Public Dataset by Olist

100,000 Orders with product, customer and reviews info

https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce?resource=download

Why the Olist Dataset Is Not Ideal

Although the **Brazilian E-Commerce Public Dataset by Olist** contains rich e-commerce information, it is not ideal for predicting **30-day customer spending** for the following reasons:

1. Customer ID Issue

Each order has a different `customer_id`, meaning the same customer may appear with multiple IDs. This makes it difficult to accurately track repeat purchases and customer spending behavior.

2. Highly Fragmented Data

The dataset is split across multiple tables (orders, payments, reviews, products, sellers), requiring complex joins and heavy preprocessing before modeling.

3. Not Focused on Spend Prediction

Many columns relate to logistics, delivery, and reviews rather than customer purchasing patterns, which are more relevant for predicting future spending.

4. High Data Engineering Effort

Significant time would be required to merge tables and construct customer-level features, which is inefficient in a hackathon environment.

UCI Machine Learning Repository

Discover datasets around the world!

https://archive.ics.uci.edu/dataset/502/online+retail+ii

Dataset Used: **Online Retail Transaction Dataset**

Dataset contains:

Column	Description
InvoiceNo	Transaction ID
StockCode	Product ID
Description	Product name
Quantity	Number of items purchased
InvoiceDate	Date and time of purchase

Why this dataset was chosen:

- Represents real retail transactions
- Contains customer identifiers
- Includes timestamps
- Transaction values can be computed
- Customers have multiple transactions

Column	Description
UnitPrice	Price per item
CustomerID	Customer identifier
Country	Customer location

Target Definition

A **cutoff date (T)** is defined.

Feature Window

Transactions before T are used to generate customer behavior features.

Prediction Window

Transactions between **T and T + 30 days** are used to compute the target variable.

Target formula:

```
future_spend_30d = sum(Quantity × UnitPrice)
within next 30 days
```

Data Preprocessing Plan

Steps:

1. Load dataset
2. Remove rows with missing `CustomerID`
3. Remove cancelled transactions
4. Convert `InvoiceDate` to datetime format
5. Create transaction value

```
transaction_value = Quantity × UnitPrice
```

Feature Engineering Plan

Customer-level features will be generated.

RFM Features

Recency - Days since last purchase

Frequency - Number of transactions

Monetary -Total spending

Exploratory Data Analysis (EDA)

EDA will be performed to understand customer purchasing patterns.

Analysis includes:

- Customer spending distribution
- Purchase frequency
- Transaction value distribution
- Country-based purchase patterns
- Feature correlation analysis

Tools used:

- Pandas
- Matplotlib
- Seaborn

Model Strategy

Baseline Model

Linear Regression

Models to Evaluate

- Linear Regression
- Random Forest Regressor
- Gradient Boosting

- XGBoost

The best model will be selected based on evaluation metrics.

Evaluation Metrics

Model performance will be evaluated using:

MAE (Mean Absolute Error) - Average prediction error.

RMSE (Root Mean Squared Error) - Penalizes larger prediction errors.

R^2 Score - Measures how well the model explains variance.

Team Role Allocation

Team Member	Responsibilities
Tharun	Dataset collection, dataset loading, data cleaning, handling missing values, removing cancelled transactions
Yokesh	Assist in dataset collection, perform Exploratory Data Analysis (EDA), visualize customer behaviour, analyze spending patterns
Yogesh	Project timeline planning, workflow coordination, and testing the model predictions
Mayank	Documentation, ML model development, data science pipeline design, API and UI integration, and project demonstration

Kaggle file